Ayush Madhav Kumar

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Education

University of Michigan, College of Engineering **Aug 2025 – Dec 2028**

B.S.E. Computer Science, Minor in Mathematics; GPA 3.60/4.00 *Ann Arbor, MI*

– **Coursework:** Data Structures and Algorithms, Computer Organization, Discrete Mathematics, Linear Algebra

Experience

NSF Pathways to Open Source Ecosystems **May 2026 – Present**

Undergraduate Researcher *Ann Arbor, MI*

– Built the benchmark suites for Jac's experimental GPU walker compiler on a $300K NSF grant, checking every output against independent oracles on an RTX 3090
– Designed a layout benchmark that runs identical PTX under random and access-order node placement, timed in Nsight Systems: up to 1.75× faster kernels on 256^3 matmul across 3 seeds
– Wrote six graph application benchmarks timing the full `run_walkers` GPU call: 29–94× over serial Jac on shared graphs, but only about 2× when each walker owns its nodes
– Built Marketing Agent Studio (55 Jac modules) and NightShift, a nightly compiler-audit agent that opens PRs

Jaseci Labs **May 2026 – Aug 2026**

Software Engineering Intern *Ann Arbor, MI*

– Fine-tuned Qwen3-Coder 30B into a coding agent for the Jac language, reaching 94% compile-pass on a held-out benchmark, using Unsloth and 4-bit QLoRA
– Generated 300k+ training examples through a synthetic data pipeline, each gated on a compiler and unit tests
– Cut the model's fallback to Python patterns 82% with a four-stage SFT, debugging, DPO, and agentic curriculum

Cactus (YC S25) **Jan 2026 – Present**

Open Source Contributor *Remote*

– Fixed image completions hanging on Metal and CPU: a shape bug split a 1,043-token prompt into 1,000 chunks
– Fixed two C++ tokenizer bugs: byte-fallback pieces decoded as literal text, and vocab lines mis-split on tabs
– Documented custom vocabulary and boost options for transcription across four SDKs: Python, Swift, Kotlin, Flutter

CareTether (Patch Summer Accelerator '25) **Jul 2025 – Aug 2025**

Co-Founder *Dublin, Ireland*

– Built the app front end for a senior-care wearable and ran a pilot with 5+ families over 8 weeks
– Reached 20,000+ people through social campaigns and pitched NGOs and care homes on distribution

Projects

NASA SUITS Challenge | ***AI Lead, team of 6; Python, PyTorch, Unity***

– Led the AI subteam on autonomous lunar EVA navigation to 96.8% waypoint accuracy, with A*/JPS planning over a 3-layer occupancy grid and 17-ray LiDAR fusion
– Trained the obstacle detection models served in real time behind a hot-swappable backend: a temporal ResidualBeamCNN and a BEV cross-attention transformer
– Built the voice pipeline at 200ms end to end, streaming 16 kHz PCM over WebSocket into faster-whisper

OSINT Geospatial Intelligence Platform | ***TypeScript, React, CesiumJS, Node.js, WebSocket***

– Onboarded any public data feed as a single YAML file with no code changes, with JSON, GeoJSON, XML and CSV parsers
– Eliminated a 372K-row duplication bug across 31K entities and 780K+ observations with a deterministic observation ID and a unique natural-key index
– Kept tens of thousands of live markers interactive on integrated GPUs, batching into one CesiumJS BillboardCollection and streaming delta-only updates

Technical Skills

- **Languages:** Python, C++, C, Java, TypeScript, JavaScript, Rust, SQL, Jac
- **GPU:** CUDA, PTX, Nsight Systems, GPU kernel benchmarking, graph memory layouts, differential testing
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace Transformers, Unsloth, LoRA/QLoRA, DPO, vLLM, ONNX Runtime, scikit-learn, RAG, quantization
- **Backend and Infrastructure:** FastAPI, Flask, Django, Node.js, Express, REST, gRPC, GraphQL, WebSockets, RabbitMQ, Docker, Kubernetes, Linux, Amazon Web Services, Terraform, GitHub Actions
- **Data and Frontend:** PostgreSQL, MySQL, MongoDB, Redis, SQLite, Pandas, NumPy, React, Next.js, React Native, Redux Toolkit, CesiumJS, Vite, Flutter